

Summary of LSTM Model Performance

The LSTM model shows the following performance:

Precision, Recall, F1-score:

Negative: Precision: 0.33, Recall: 0.33, F1-score: 0.33

Neutral: Precision: 0.24, Recall: 0.35, F1-score: 0.29

Positive: Precision: 0.90, Recall: 0.85, F1-score: 0.88

Overall Accuracy: 0.79

Interpretation of Confusion Matrix:

The confusion matrix for the LSTM model (shown above) indicates:

The model correctly identified 1 out of 3 'Negative' reviews, and misclassified 1 as 'Neutral' and 1 as 'Positive'.

For 'Neutral' reviews, 12 were correctly classified, 1 was misclassified as 'Negative', and 22 were misclassified as 'Positive'.

The model performed reasonably well on 'Positive' reviews, correctly classifying 218 out of 256, with 36 misclassified as 'Neutral' and 2 as 'Negative'.

Similar to Logistic Regression, the LSTM model also struggles with the minority classes ('Negative' and 'Neutral'), with lower precision and recall for these categories. While its overall accuracy is slightly lower than Logistic Regression, it shows a similar pattern of misclassifications, particularly confusing 'Neutral' with 'Positive' reviews.

